

Two-way fixed effects estimators with heterogeneous treatment effects*

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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1 Introduction

A popular method to estimate the effect of a treatment on an outcome is to compare over time groups experiencing different evolutions of their exposure to treatment. In practice, this idea is implemented by estimating regressions that control for group and time fixed effects. Hereafter, we refer to those as two-way fixed effects (FE) regressions. We conducted a survey, and found that 20% of all empirical articles published by the American Economic Review (AER) between 2010 and 2012 have used a two-way FE regression to estimate the effect of a treatment on an outcome. When the treatment effect is constant across groups and over time, such regressions estimate that effect under the standard “common trends” assumption. However, it is often implausible that the treatment effect is constant. For instance, the minimum wage’s effect on employment may vary across US counties, and may change over time. This paper examines the properties of two-way FE regressions when the constant effect assumption is violated.

We start by assuming that all observations in the same (g, t) cell have the same treatment and that the treatment is binary, as is for instance the case when the treatment is a county-level law. We consider the regression of $Y_{i,g,t}$, the outcome of unit i in group g at period t on group fixed effects, period fixed effects, and $D_{g,t}$, the treatment in group g at period t . Let $\widehat{\beta}_{fe}$ denote the coefficient of $D_{g,t}$, and let β_{fe} denote its expectation. Under the common trends assumption, we show that β_{fe} is equal to a weighted sum of the treatment effect in each treated (g, t) cell:

$$\beta_{fe} = E \left(\sum_{(g,t): D_{g,t}=1} W_{g,t} \Delta_{g,t} \right). \quad (1)$$

$\Delta_{g,t}$ is the average treatment effect (ATE) in group g and period t and the weights $W_{g,t}$ s sum to one but may be negative. Negative weights arise because $\widehat{\beta}_{fe}$ is a weighted sum of several difference-in-differences (DID), which compare the evolution of the outcome between consecutive time periods across pairs of groups. However, the “control group” in some of those comparisons may be treated at both periods. Then, its treatment effect at the second period gets differenced out by the DID, hence the negative weights.

The negative weights are an issue when the ATEs are heterogeneous across groups or periods. Then, one could have that β_{fe} is negative while all the ATEs are positive. For instance, $1.5 \times 1 - 0.5 \times 4$, a weighted sum of 1 and 4, is strictly negative. Using the data set of Gentzkow et al. (2011), we find that 40% of the weights attached to β_{fe} are negative, so β_{fe} is not robust to heterogeneous effects.¹

Researchers may want to know how serious that issue is in the application they consider. We show that conditional on all treatments, the absolute value of the expectation of $\widehat{\beta}_{fe}$ divided by

¹ Gentzkow et al. (2011) do not estimate β_{fe} , but β_{fd} , the treatment coefficient in the first-difference regression defined below. 46% of the weights attached to β_{fd} are strictly negative.

Our paper is related to the DID literature. Our main result generalizes Theorem 1 in de Chaisemartin and D’Haultfoeuille (2018). When the data has two groups and two periods, the Wald-DID estimand considered therein is equal to β_{fe} and β_{fd} . Our results on β_{fe} and β_{fd} are thus extensions of that theorem to the case with multiple periods and groups.³ Moreover, our DID_M estimator is related to the Wald-TC estimator with many groups and periods proposed in de Chaisemartin and D’Haultfoeuille (2018), and to the multi-period DID estimator proposed by Imai and Kim (2018). In Section 4, we explain the differences between those three estimators.

More recently, Borusyak and Jaravel (2017), Abraham and Sun (2018), Athey and Imbens (2018), Callaway and Sant’Anna (2018), and Goodman-Bacon (2018) study the special case of staggered adoption designs, where the treatment of a group is weakly increasing over time. Those papers derive some important results specific to that design that we do not consider here. Still, some of the results in those papers are related to ours, and we describe precisely those connections later in the paper. The most important dimension on which our paper differs from those is that our results apply to any two-way fixed effects regressions, not only to those with staggered adoption. In our survey of the AER papers estimating two-way fixed effects regressions, less than 10% have a staggered adoption design. This suggests that while staggered adoptions are an important research design, they may account for a relatively small minority of the applications where two-way fixed effects regressions have been used.

The paper is organized as follows. Section 2 introduces the set-up. Section 3 presents our decomposition results. Section 4 introduces our alternative estimator. Section 5 briefly describes some of the extensions covered in our Web Appendix. Section 6 presents our survey of the articles published in the AER, and our two empirical applications.

2 Set up

One considers observations that can be divided into G groups and T periods. For every $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $N_{g,t}$ denote the number of observations in group g at period t , and let $N = \sum_{g,t} N_{g,t}$ be the total number of observations. The data may be an individual-level panel or repeated cross-section data set where groups are, say, individuals’ county of birth. The data could also be a cross-section where cohort of birth plays the role of time. For instance, Duflo (2001) compares the schooling of different cohorts in Indonesia, some of which were exposed to a school construction program. It is also possible that for all (g, t) , $N_{g,t} = 1$, e.g. a group is one individual or firm. All of the above are special cases of the data structure we consider.

One is interested in measuring the effect of a treatment on some outcome. Throughout the paper

³In fact, a preliminary version of our main result appeared in a working paper version of de Chaisemartin and D’Haultfoeuille (2018) (see Theorems S1 and S2 in de Chaisemartin and D’Haultfoeuille, 2015).

Assumption 4 (*Strong exogeneity*) For all $(g, t) \in \{1, \dots, G\} \times \{2, \dots, T\}$,
 $E(Y_{g,t}(0) - Y_{g,t-1}(0) | D_{g,1}, \dots, D_{g,T}) = E(Y_{g,t}(0) - Y_{g,t-1}(0)).$

Assumption 4 requires that the shocks affecting a group's $Y_{g,t}(0)$ be mean independent of that group's treatment sequence. This rules out the possibility that a group gets treated because it experiences negative shocks, the so-called Ashenfelter's dip (see Ashenfelter, 1978). Assumption 4 is related to the strong exogeneity condition in panel data models, which, as is well-known, is necessary to obtain the consistency of the fixed effects estimator (see, e.g., Wooldridge, 2002).

We now define the FE regression described in the introduction.⁴

Regression 1 (*Fixed-effects regression*)

Let $\hat{\beta}_{fe}$ denote the coefficient of $D_{g,t}$ in an OLS regression of $Y_{i,g,t}$ on group fixed effects, period fixed effects, and $D_{g,t}$. Let $\beta_{fe} = E[\hat{\beta}_{fe}]$.⁵

For all g and t , let $N_{g,\cdot} = \sum_{t=1}^T N_{g,t}$ and $N_{\cdot,t} = \sum_{g=1}^G N_{g,t}$ respectively denote the total number of observations in group g and in period t . For any variable $X_{g,t}$ defined in each (g, t) cell, let $X_{g,\cdot} = \sum_{t=1}^T (N_{g,t}/N_{g,\cdot}) X_{g,t}$ denote the average value of $X_{g,t}$ in group g , let $X_{\cdot,t} = \sum_{g=1}^G (N_{g,t}/N_{\cdot,t}) X_{g,t}$ denote the average value of $X_{g,t}$ in period t , and let $X_{\cdot,\cdot} = \sum_{g,t} (N_{g,t}/N) X_{g,t}$ denote the average value of $X_{g,t}$. For instance, $D_{3,\cdot}$ and $D_{\cdot,2}$ respectively denote the average treatment in group 3 across time and in period 2 across groups, whereas $Y_{\cdot,\cdot}$ denotes the average value of the outcome across groups and time. Finally, for any variable $X_{g,t}$, we let $\mathbf{X}$ denote the vector $(X_{g,t})_{(g,t) \in \{1, \dots, G\} \times \{1, \dots, T\}}$ collecting the values of that variable in each (g, t) cell. For instance, $\mathbf{D}$ is the vector $(D_{g,t})_{(g,t) \in \{1, \dots, G\} \times \{1, \dots, T\}}$ collecting the treatments of all the (g, t) cells.

3 Two-way fixed effects regressions

3.1 A decomposition result

We study the FE regression under the following common trends assumption.

Assumption 5 (*Common trends*) For $t \geq 2$, $E(Y_{g,t}(0) - Y_{g,t-1}(0))$ does not vary across g .

Assumption 5 requires that the expectation of the outcome without treatment follow the same evolution over time in every group. When t represents birth cohorts, Assumption 5 requires that the outcome difference between consecutive cohorts be the same across groups.

⁴ Throughout the paper, we assume that $D_{g,t}$ in Regression 1 and $D_{g,t} - D_{g,t-1}$ in Regression 2 below are not collinear with the other independent variables in those regressions, so $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ are well-defined.

⁵ As the independent variables in Regression 1 are constant within each (g, t) cell, Regression 1 is equivalent to a (g, t) -level regression of $Y_{g,t}$ on group and period fixed effects and $D_{g,t}$, weighted by $N_{g,t}$.

Let $N_1 = \sum_{i,g,t} D_{i,g,t}$ denote the number of treated units, let

$$\Delta^{TR} = \frac{1}{N_1} \sum_{(i,g,t): D_{i,g,t}=1} [Y_{i,g,t}(1) - Y_{i,g,t}(0)]$$

denote the average treatment effect across all treated units, and let $\delta^{TR} = E[\Delta^{TR}]$ denote the expectation of that parameter, hereafter referred to as the ATT. For any $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let

$$\Delta_{g,t} = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} [Y_{i,g,t}(1) - Y_{i,g,t}(0)]$$

denote the ATE in cell (g, t) . δ^{TR} is equal to the expectation of a weighted average of the treated cells' $\Delta_{g,t}$ s:

$$\delta^{TR} = E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} \Delta_{g,t} \right]. \quad (2)$$

Under the common trends assumption, we show that β_{fe} is also equal to the expectation of a weighted sum of the $\Delta_{g,t}$ s, with potentially some negative weights.

Let $\varepsilon_{g,t}$ denote the residual of observations in cell (g, t) in the regression of $D_{g,t}$ on group and period fixed effects:⁶

$$D_{g,t} = \alpha + \gamma_g + \lambda_t + \varepsilon_{g,t}.$$

One can show that if the regressors in Regression 1 are not collinear, the average value of $\varepsilon_{g,t}$ across all treated (g, t) cells differs from 0: $\sum_{(g,t): D_{g,t}=1} (N_{g,t}/N_1) \varepsilon_{g,t} \neq 0$. Then we let $w_{g,t}$ denote $\varepsilon_{g,t}$ divided by that average:

$$w_{g,t} = \frac{\varepsilon_{g,t}}{\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} \varepsilon_{g,t}}.$$

Theorem 1 *Suppose that Assumptions 1-5 hold. Then,*⁷

$$\beta_{fe} = E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \Delta_{g,t} \right].$$

⁶ $\varepsilon_{g,t}$ arises from a unit-level regression, where the dependent and independent variables only vary at the (g, t) level. Therefore, all the units in the same (g, t) cell have the same value of $\varepsilon_{g,t}$.

⁷ In the proof, we show the following, stronger result:

$$E[\hat{\beta}_{fe} | \mathbf{D}] = \sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} E[\Delta_{g,t} | \mathbf{D}].$$

$t - 1$ and t would have experienced if they had remained treated. In staggered adoption designs, Assumption 9 and 10 are not necessary for identification, because no group leaves the treatment. Together, Assumptions 5 and 10 imply that the ATE follows the same evolution over time in every group: $E(\Delta_{g,t}) = \eta_t + \theta_g$.¹³ This still allows for heterogeneous treatment effects across groups and over time.¹⁴

Assumption 11 (*Existence of “stable” groups*) For all $t \in \{2, \dots, T\}$:

1. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 0$.
2. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 1, D_{g,t} = 0$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 1$.

The first point of the stable groups assumption requires that between each pair of consecutive time periods, if there is a “joiner” (i.e., a group switching from being untreated to treated), then there should be another group that is untreated at both dates. The second point requires that between each pair of consecutive time periods, if there is a “leaver” (i.e., a group switching from being treated to untreated), then there should be another group that is treated at both dates.

Notice that under Assumption 11, groups’ treatments are not independent, so Assumption 3 cannot hold. Accordingly, we replace Assumption 3 by Assumption 12 below. Assumption 12 requires that conditional on its own treatments, a group’s outcomes be mean independent of the other groups’ treatments. It is weaker than Assumption 3. Assumption 11 is necessary to show that our estimator is unbiased, but it is not necessary to show that it is consistent. Accordingly, in Section ?? of the Web Appendix, we show that our estimator is consistent under Assumption 3. For every $g \in \{1, \dots, G\}$, let $\mathbf{D}_g = (D_{1,g}, \dots, D_{T,g})$.

Assumption 12 (*Mean independence between a group’s outcome and other groups treatments*) For all g and t , $E(Y_{g,t}(0)|\mathbf{D}) = E(Y_{g,t}(0)|\mathbf{D}_g)$ and $E(Y_{g,t}(1)|\mathbf{D}) = E(Y_{g,t}(1)|\mathbf{D}_g)$.

We can now define our estimator. For all $t \in \{2, \dots, T\}$ and for all $(d, d') \in \{0, 1\}^2$, let

$$N_{d,d',t} = \sum_{g: D_{g,t}=d, D_{g,t-1}=d'} N_{g,t} \quad (3)$$

¹³ It should be possible to weaken Assumptions 9-10, in particular to account for dynamic effects where $\Delta_{g,t}$ may depend on $(D_{g,1}, \dots, D_{g,t-1})$. This would however introduce complications that are beyond the scope of this paper.

¹⁴ Imposing Assumptions 9 and 10 does not change the decompositions obtained in Theorems 1 and 2. $Y_{g,t}(1)$ is observed for all the treated (g, t) cells entering these decompositions, so those assumptions do not bring identifying information for those cells.

denote the number of observations with treatment d' at period $t - 1$ and d at period t . Let

$$\begin{aligned} \text{DID}_{+,t} &= \sum_{g:D_{g,t}=1, D_{g,t-1}=0} \frac{N_{g,t}}{N_{1,0,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=D_{g,t-1}=0} \frac{N_{g,t}}{N_{0,0,t}} (Y_{g,t} - Y_{g,t-1}), \\ \text{DID}_{-,t} &= \sum_{g:D_{g,t}=D_{g,t-1}=1} \frac{N_{g,t}}{N_{1,1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=0, D_{g,t-1}=1} \frac{N_{g,t}}{N_{0,1,t}} (Y_{g,t} - Y_{g,t-1}). \end{aligned}$$

Note that $\text{DID}_{+,t}$ is not defined when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 0$, or no group such that $D_{g,t} = 0, D_{g,t-1} = 0$. In such instances, we let $\text{DID}_{+,t} = 0$. Similarly, let $\text{DID}_{-,t} = 0$ when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 1$ or no group such that $D_{g,t} = 0, D_{g,t-1} = 1$. Finally, let

$$\text{DID}_M = \sum_{t=2}^T \left(\frac{N_{1,0,t}}{N_S} \text{DID}_{+,t} + \frac{N_{0,1,t}}{N_S} \text{DID}_{-,t} \right).$$

Theorem 3 *If Assumptions 1, 2, 4, 5, and 9-12 hold, $E[\text{DID}_M] = \delta^S$.*

In Section ?? of the Web Appendix, we also show that when G goes to infinity, DID_M is a consistent and asymptotically normal estimator of δ^S . The DID_M estimator is computed by the `fuzzydid` and `did_multipligt` Stata packages.

Here is the intuition underlying Theorem 3. $\text{DID}_{+,t}$ compares the evolution of the mean outcome between $t - 1$ and t in two sets of groups: the joiners, and those remaining untreated. Under Assumptions 4 and 5, $\text{DID}_{+,t}$ estimates the joiners' treatment effect. Similarly, $\text{DID}_{-,t}$ compares the evolution of the outcome between $t - 1$ and t in two sets of groups: those remaining treated, and the leavers. Under Assumptions 9 and 10, it estimates the leavers' treatment effect. Finally, DID_M is a weighted average of those DIDs. Note that in staggered designs, there are no groups whose treatment decreases over time, so DID_M is only a weighted average of the $\text{DID}_{+,t}$ estimators. Note also that one can separately estimate the joiners' and the leavers' treatment effect, by computing separately weighted averages of the $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ estimators. The former estimator only relies on Assumptions 4 and 5, while the latter only relies on Assumptions 9 and 10.

DID_M is related to two other estimators. First, it is related to the Wald-TC estimator in point 2 of Theorem S1 in the Web Appendix of de Chaisemartin and D'Haultfœuille (2018), but the weighting of $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ therein differs. As a result, DID_M estimates Δ^S under weaker assumptions. DID_M is also related to the multi-period DID estimator in Imai and Kim (2018). However, the multi-period DID estimator is a weighted average of the $\text{DID}_{+,t}$, so it does not estimate the leavers' treatment effect, and applies to a smaller population. Besides, Imai and Kim (2018) do not establish the properties of their estimator. Finally, they do not generalize it to non-binary treatments, something we do in Section ?? of the Web Appendix.